Nyenje Bay Wind/Waves Analysis Tool

Technical Framework Report

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Location: **Nyenje Bay, Lake Kariba**

Coordinates: **16.53S, 28.83E**

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Engineering Design Basis for Floating Photovoltaic (FPV) Structures

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Executive Summary

This report documents the technical framework of the Nyenje Bay Wind/Waves Analysis Tool. The tool integrates multiple layers:

- **User Interface Layer:** Streamlit-based web dashboard
- **Scientific Computing Layer:** NumPy, SciPy, Pandas, Xarray
- **Data Layer:** ERA5 reanalysis, CCMP satellite, ground measurements
- **Wave Modeling Layer:** SWAN model integration

Key Scientific Calculations:

- Wind averaging periods: 3-minute, 10-minute, 3-second gust, 10-second gust
- Gust factors: 1.15 (3-min), 1.45 (3-sec), 1.35 (10-sec)
- JONSWAP wave height formula
- Gumbel extreme value analysis for return periods

1 Tool Architecture

1.1 Four-Layer Framework

The tool is built on a four-layer architecture:

Layer	**Technology**	**Functions**
User Interface	Streamlit	Web dashboard, interactive plots
Scientific Computing	NumPy, SciPy	Wind averaging, wave calculations
Data Processing	Xarray, Pandas	ERA5 reading, grid interpolation
External Models	SWAN	Wave distribution

Table 1: Architecture Layers

2 Wind Averaging Calculations

2.1 Overview

The tool calculates five different wind averaging periods:

Period	**Gust Factor**	**Application**
Hourly Mean	1.00	Reference wind speed
3-Minute Average	1.15	Dynamic response
10-Minute Average	1.00	Marine forecast
3-Second Gust	1.45	Ultimate Limit State (ULS)
10-Second Gust	1.35	Serviceability Limit State (SLS)

Table 2: Wind Averaging Periods

2.2 Exponential Weighted Moving Average

For hourly data, the tool uses Exponential Weighted Moving Average:

$$\mathrm{EWMA}_t = \alpha \cdot x_t + (1 - \alpha) \cdot \mathrm{EWMA}_{t-1} \tag{1}$$

Where:

- $\alpha = \frac{2}{\text{span}+1}$ is the smoothing factor
- For 3-minute average: span = 3, $\alpha = 0.5$
- For 10-minute average: span = 10, $\alpha \approx 0.18$

2.3 Gust Factor Formula

$$\text{Gust Factor} = \frac{\text{Peak Gust Speed}}{\text{Mean Wind Speed}} \tag{2}$$

Period	**Gust Factor**
3-second gust	1.45
10-second gust	1.35
3-minute average	1.15

Table 3: Gust Factor Derivation

3 Wave Height Calculation

3.1 JONSWAP Formula

The Joint North Sea Wave Project (JONSWAP) formula is used for fetch-limited waves.

3.1.1 Dimensionless Fetch

$$\tilde{x} = \frac{g \cdot F}{U^2} \tag{3}$$

Where:

- $g = 9.81$ m/s^2 (gravity)
- F = fetch length (meters)
- U = wind speed (m/s)

3.1.2 Significant Wave Height

$$H_s = 0.0016 \cdot \frac{U^2}{g} \cdot \sqrt{\tilde{x}} \tag{4}$$

3.1.3 Peak Wave Period

$$T_p = 0.286 \cdot \frac{U}{g} \cdot (\tilde{x})^{0.33} \tag{5}$$

3.2 Simplified Formula

$$\boxed{H_s = 0.0016 \cdot U \cdot \sqrt{g \cdot F}} \tag{6}$$

3.3 Example Calculations

Wind Speed (m/s)	Wave Height Hs (m)	Period Tp (s)
5	0.07	1.2
8	0.18	2.0
10	0.29	2.3
12	0.41	2.7
15	0.64	3.2
18	0.92	3.7
20	1.14	4.0

Table 4: Wave Height for Different Wind Speeds (Fetch = 4 km)

4 Extreme Value Analysis

4.1 Gumbel Distribution

The cumulative distribution function (CDF) is:

$$F(x) = \exp\left(-\exp\left(-\frac{x-\mu}{\beta}\right)\right) \tag{7}$$

Where:

- μ = location parameter
- β = scale parameter

4.2 Return Period Calculation

For return period T (years):

$$x(T) = \mu - \beta \cdot \ln\left(-\ln\left(1 - \frac{1}{T}\right)\right) \tag{8}$$

4.3 Results for Nyenje Bay

Return Period (years)	**Wind Speed (m/s)**	**Wave Height (m)**
2	11.2	0.85
5	12.8	1.02
10	14.0	1.15
20	15.2	1.28
25	15.6	1.32
50	16.8	1.48
100	18.0	1.62

Table 5: Return Period Wave Heights

5 Validation Results

5.1 Kariba Airport Comparison

Metric	Raw ERA5	Debiased ERA5	Improvement
Correlation (r)	0.39	0.502	+29%
Bias (m/s)	+2.04	+0.01	99.5%
RMSE (m/s)	2.45	0.52	79%

Table 6: Validation Statistics (2001-2005)

5.2 Interpretation of r = 0.502

$$r = 0.502 \quad \Rightarrow \quad R^2 = 0.252 \tag{9}$$

This means:

- 25.2% of wind variance is shared between locations
- 74.8% is due to local effects (expected for 3 km separation)
- The correlation exceeds the expected value of 0.40

6 Technology Stack

Component	**Technology**	**Purpose**
UI Framework	Streamlit	Web dashboard
Scientific	NumPy	Array operations
Data Analysis	Pandas	DataFrames
NetCDF	Xarray	ERA5 reading
Statistics	SciPy	Gumbel distribution
Visualization	Plotly	Interactive plots
Wave Model	SWAN	Wave distribution

Table 7: Technology Stack

7 Conclusions

7.1 Key Technical Achievements

1. **Multi-Layer Architecture**: Separation of UI, computation, and data
2. **Wind Averaging**: EWMA method for 3-min and 10-min averages
3. **Gust Factors**: Engineering-standard factors (1.45, 1.35)
4. **Wave Calculations**: JONSWAP fetch-limited formula
5. **Extreme Analysis**: Gumbel distribution for return periods
6. **Validation**: $r = 0.502$ correlation with ground measurements

7.2 Framework Independence

The scientific core is completely independent of the user interface:

- Same calculations work in command line, web app, or API
- Streamlit is only for visualization, not computation
- Core functions can be reused in other projects